

BundleWise: A Retail Analytics Tool for Smart Bundling and Profit-Driven Pricing

Problem Statement	Solution Summary and Customer	Key Differentiators	Architecture and Technology
Many small and medium-sized retailers struggle to identify the right product bundles and set competitive prices due to scattered sales data, a lack of forecasting tools, and limited technical expertise. This often leads to missed sales opportunities, poor inventory management, and reduced profits. Existing systems typically fail to account for changing customer behavior, seasonal trends, or dynamic pricing needs. There is a need for a simple, web-based solution that helps retailers analyze sales data, discover frequently bought product combinations, forecast future demand, and optimize pricing strategies. The BundleWise project addresses this by offering an easy-to-use platform that combines pattern mining, predictive analytics, and pricing recommendations to support better decision-making and improve business outcomes.	Solution Summary BundleWise is a web-based retail analytics tool designed to help small and medium-sized retailers make smarter bundling and pricing decisions. It enables users to upload their sales data in CSV/XLSX format and automatically processes it to identify frequently purchased product combinations using sequential pattern mining. The system also forecasts future sales and profit trends using time series models like ARIMA, Prophet, or LSTM, and recommends optimal pricing strategies based on demand and seasonal patterns. With its interactive dashboard and easy-to-use interface, BundleWise empowers retailers to gain data-driven insights without requiring technical expertise, ultimately helping them improve profitability and customer satisfaction. Customer • Small and medium-sized retailers	• Time-Aware Bundling: Finds product sequences across visits, not just single transactions. • Forecast-Driven Pricing: Uses sales forecasts to suggest optimized, seasonal pricing. • Non-Technical Friendly: Easy-to-use dashboard, no coding required. • POS-Free Operation: Works via CSV uploads—no system integration needed. • Handles Messy Data: Cleans and analyzes even incomplete or scattered sales data. • Cloud-Based & Lightweight: Accessible online with no setup or installation.	• 3-Tier Architecture • Client-Server Layered Design • Pattern Mining & Forecasting Models • FastAPI / Flask REST API • Python-Based Analytics Engine • HTML, CSS, Plotly Dash Components • CSV File Upload & Report Export • Cloud Deployment
Scaling Opportunities	Support Needed	Contacts and References	
• BundleWise is built for scalable growth in users and data. • Supports vertical and horizontal scaling in cloud environments. • It has the potential to become a full commercial retail web app. • Future upgrades may include real-time POS/ERP integration.	• Technical Support: Maintain backend, APIs, and update models as needed. • User Training: Provide simple guides and support for retail users. • Cloud Infrastructure: Ensure smooth, scalable cloud deployment. • Security Compliance: Implement data privacy and GDPR practices. • UI/UX Design: Enhance dashboard usability and mobile responsiveness. • Marketing & Outreach: Promote BundleWise and collect user feedback.	Students Details • 2448005: Abishai Tom • 2448033: Liya Khatija • 2448036: Mansi Sapariya Faculty Details • Dr. Apash Roy, Project Guide	